SARATH SATHEESH

M.Eng. candidate, Mechatronics & Cyber-Physical Systems

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SUMMARY

I am finishing my M.Eng. in Mechatronics and Cyber-Physical Systems at TH Deggendorf (all coursework done, thesis next), after a B.Tech. in Mechanical Engineering. During my studies I built a VR driving simulator in Unity as part of a team, wrote a gear generator in Lua that we 3D-printed, and did simulation work in Unity. I program in Python and C++, use SolidWorks for CAD, and am learning ROS 2. I am looking for a thesis, internship or Werkstudent role in robotics, mechatronics or automation and can start immediately.

EDUCATION

M.Eng. Mechatronics and Cyber-Physical Systems | *THD, Deggendorf, Germany* **Oct 2023 – present**

All coursework is complete; thesis and seminar outstanding. Modules include Simulation Systems for Advanced Applications, Mechatronic System Simulation, Cooperative and Autonomous Systems, Cyber-Physical Systems, Functional Safety and Additive Manufacturing.

B.Tech. Mechanical Engineering | *APJ Abdul Kalam Technological University, Kerala, India* **Jun 2018 – Jun 2022**

Key modules: CAD & CAE, Material Science, Manufacturing Technology, Fluid & Thermal Engineering.

SKILLS

- **Programming:** Python, C++, Lua
- **Modelling & simulation:** MATLAB, Unity, MuJoCo (basic)
- **Robotics:** ROS 2 (basics; Udemy course in progress), ROS
- **Design & manufacturing:** SolidWorks, AutoCAD, IceSL, FFF 3D printing
- **Tools:** Git, Linux, MS Office

PROJECTS

Driver Perspective Simulator: Active Safety in VR | *THD, team project* **2025**

- A driving simulator you experience with a VR headset, built in Unity (a game engine) by a university team.
- You sit in the driver's seat in dangerous traffic situations from the Euro NCAP car safety tests, such as a pedestrian crossing the road. The car's automatic emergency braking (AEB) has to react by itself.
- My part: the scenario logic (what happens in each situation), the car's movement and path, and connecting the headset, the simulation and the braking system so they worked together.
- I ran the scenarios, checked when the braking reacted, and kept the documentation so others can reuse the simulator.

Involute Gear: Effect of Profile Shift | *THD, team project* **2024**

- A university case study about gears, done by a team of three.
- When a gear is made, the cutting tool can be moved slightly (profile shift). It keeps small gears strong and lets two gears run at a different distance apart.
- We wrote a program in IceSL (a 3D-printing program that uses the Lua language) that draws the gears from a few numbers: teeth, tooth size, pressure angle, width and profile shift.
- We printed the gears, shaft pins and a stand and assembled them into a demo model with a handle.
- My part: I co-wrote the technical report on the gear geometry and the user guide for the program.

Capacitive Fuel Level Sensor (concept study) **Bachelor project**

- A capacitive sensor measures how much fuel is around it through a change in an electrical value (capacitance).
- This was a concept study and nothing was built. I worked out how the sensor works, what electronics would read it, and how Bluetooth could send the reading, and I presented it.

Heat Transfer Enhancement in a Double Pipe Heat Exchanger **Bachelor project**

- A double pipe heat exchanger passes heat from a hot liquid in one pipe to a cold liquid in the other.
- I planned the experiments and analyzed the results to see which design changes moved more heat.

EXPERIENCE

Junior Automation Engineer | *Logix Space, Pandalam, India* **Jul 2022 – Aug 2023**

- Helped the team build prototypes and test setups for automation systems: assembly, sensor and actuator integration, calibration and testing.
- Watched commissioning on the workshop floor and helped trace hardware and sensor faults.
- Kept test records and documentation, working with mechanical, electrical and software colleagues.

Crew Member (part-time) | *McDonald's, Regensburg* **2023 – present**

- Part-time job next to my studies in a fast-paced team. I use German every day.

CERTIFICATES

MATLAB Onramp (MathWorks) · ROS: Hello (Real) World (edX) · Tele-Robotics (vhb) · Unity: Game Development & Simulation · AutoCAD 2D & 3D · Automation & Robotics Internship, Logix Space (2022) · ROS 2 course (Udemy, in progress)

LANGUAGES

English: professional working proficiency · German: A2, actively improving

OUTSIDE WORK

I build small games in Unity in my spare time and enjoy taking apart how open-world games are designed.